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# =====
# DASHBOARD VISUAL PROFISSIONAL
# =====

import numpy as np
import matplotlib.pyplot as plt

plt.style.use("dark_background")

# =====
# DADOS EXEMPLO
# =====

t = np.linspace(0, 30, 1000)

voltage = 24 - 24*np.exp(-t/3) + 0.5*np.sin(8*t)

current = 4 + 2*np.sin(2*t)*np.exp(-t/15)

power = voltage * current

eff = 92 + 4*np.sin(t/8)

# =====
# FIGURA PRINCIPAL
# =====

fig = plt.figure(figsize=(18,10))

# GRID MAIS MODERNO
grid = fig.add_gridspec(
    3,
    4,
    hspace=0.35,
    wspace=0.25
)

# =====
# TÍTULO PRINCIPAL
# =====

fig.suptitle(
    "PAINEL DE CONTROLE DO SISTEMA ELÉTRICO",

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        fontsize=26,
        color='cyan',
        fontweight='bold'
    )

# =====
# GRÁFICO PRINCIPAL GRANDE
# =====

ax_main = fig.add_subplot(grid[0:2, 0:3])

ax_main.plot(
    t,
    voltage,
    color='#00F5FF',
    linewidth=3
)

ax_main.fill_between(
    t,
    voltage,
    alpha=0.15,
    color='#00F5FF'
)

# Área de estabilidade
ax_main.axhspan(
    23,
    25,
    color='green',
    alpha=0.08
)

ax_main.set_title(
    "Estabilização da Tensão",
    fontsize=20
)

ax_main.set_ylabel(
    "Tensão (V)",
    fontsize=14
)

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ax_main.grid(
    linestyle=':',
    alpha=0.25
)

# Marcador do ponto final
ax_main.scatter(
    t[-1],
    voltage[-1],
    s=120,
    color='lime',
    zorder=5
)

ax_main.text(
    t[-1]-4,
    voltage[-1]+1,
    f"{voltage[-1]:.1f}V",
    fontsize=12,
    color='white'
)

# =====
# CORRENTE
# =====

ax2 = fig.add_subplot(grid[0,3])

ax2.plot(
    t,
    current,
    color='#FFD700',
    linewidth=2.5
)

ax2.fill_between(
    t,
    current,
    alpha=0.2,
    color='#FFD700'
)

ax2.set_title("Corrente")

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ax2.grid(alpha=0.2)

# =====
# EFICIÊNCIA
# =====

ax3 = fig.add_subplot(grid[1,3])

ax3.plot(
    t,
    eff,
    color='#00FF88',
    linewidth=2.5
)

ax3.set_ylim(80,100)

ax3.fill_between(
    t,
    eff,
    alpha=0.2,
    color='#00FF88'
)

ax3.set_title("Eficiência")

ax3.grid(alpha=0.2)

# =====
# POTÊNCIA FULL WIDTH
# =====

ax4 = fig.add_subplot(grid[2,:])

ax4.plot(
    t,
    power,
    color='#FF4D4D',
    linewidth=3
)

# Efeito glow fake

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for lw, alpha in zip([10,8,6], [0.03,0.05,0.08]):

    ax4.plot(
        t,
        power,
        color='#FF0000',
        linewidth=lw,
        alpha=alpha
    )

ax4.set_title(
    "Fluxo de Potência do Sistema",
    fontsize=18
)

ax4.set_xlabel(
    "Tempo",
    fontsize=13
)

ax4.set_ylabel(
    "Potência (W) ",
    fontsize=13
)

ax4.grid(
    linestyle=':',
    alpha=0.2
)

# =====
# REMOVE POLUIÇÃO VISUAL
# =====

for ax in fig.axes:

    ax.spines['top'].set_visible(False)
    ax.spines['right'].set_visible(False)

# =====
# MOSTRAR
# =====

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plt.show()
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